COMPUTER NUMBER SYSTEMS

Information &
Communication Technology

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ATTENDANCE

Raise your hand or say “present”

“SECRET CODE”

01001001 01001100 01010101

BY THE END OF THE LESSON, THE STUDENTS WILL BE ABLE TO:

- Identify and define the four major computer number systems.
- Differentiate the number systems based on their bases, allowable digits, and practical use in computing.
- Appreciate the importance of accuracy and precision when working with computer number systems to avoid technical errors.

FOUR NUMBER SYSTEMS

01 Decimal
(Base 10)

03 Octal
(Base 8)

02 Binary
(Based 2)

04 Hexadecimal
(Based 16)

Decimal (Base 10)

- Digits 0-9
- Used by humans for everyday use

0 1 2 3 4

5 6 7 8 9

Binary (Base 2)

- Digits 1s and 0s.
- The fundamental language of computer processors.
- ON/OFF switches

1 = ON

0 = OFF

Octal (Base 8)

- Digits 0-7.
- Historically used in computing to group binary digits.

0 1 2 3

4 5 6 7

Hexadecimal (Base 16)

- Digits 0-9 and letters A-F.
- Used in memory addresses, color codes, and MAC addresses.

0 1 2 3 4 5 6 7 8 9

A B C D E F

10 11 12 13 14 15

GROUP CHALLEGE

“Decode the device”

- Read the scenario
- Identify what number system is used in the scenario.

SCENARIO 1:

A smartphone screen pixel receives an electrical signal read as 1011 to turn specific sub-pixels ON and OFF to display a color.

SCENARIO 2:

A supermarket cashier scans a product barcode containing the standard UPC number 1234567890.

SCENARIO 3:

A computer technician troubleshooting a malfunctioning PC sees a system memory error message pop up as Error 0xF4.

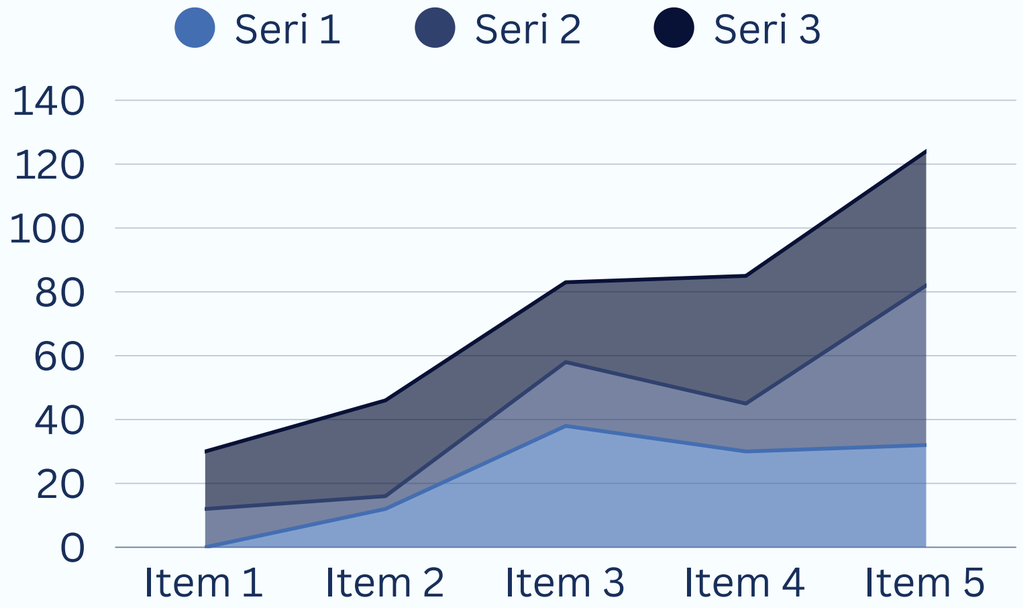
SCENARIO 4:

A Wi-Fi router's status indicator light blinks in a pattern of 1 (solid light) and 0 (no light) to show internet connectivity status.

GROUP SHARE-BACK

SUBJECT 02

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Praesent rutrum maximus mauris sed sodales. Ut rhoncus lacinia nisi eu tempus. Proin justo eros, mollis laoreet massa non, tincidunt pharetra leo. Cras facilisis leo non nibh congue volutpat. Fusce vitae sagittis dolor. Sed magna dolor, molestie vel libero id, tempus malesuada tortor.



QUIZ TIME!

Read each question carefully. Write the letter of the correct answer on your paper. For the last item, write your brief explanation clearly.

1. **Which of the following number systems is exclusively understood by computer hardware processors using electronic states of ON and OFF?**

A. Decimal

B. Binary

C. Octal

D. Hexadecimal

2. **Why do computer technicians and programmers use Hexadecimal (Base 16) instead of writing long strings of binary numbers (Base 2)?**

A. Because computers cannot read binary anymore.
B. Because hexadecimal is much shorter and acts as a convenient shorthand for binary data.
C. Because humans only have 16 fingers.
D. Because it only uses the digits 0 and 1.

3. **Identify the base and allowable digits of the Octal number system.**

A. Base 10, digits 0 to 9

B. Base 2, digits 0 and 1

C. Base 8, digits 0 to 7

D. Base 16, digits 0 to 9 and A to F

4. **If a web designer encounters a color code written as #FF5733 in Canva or Photoshop, which computer number system is being utilized?**

A. Decimal
B. Binary
C. Octal
D. Hexadecimal

5. **Short Answer: In your own words, differentiate the Decimal system from the Binary system in terms of who uses them and why.**

THANK YOU